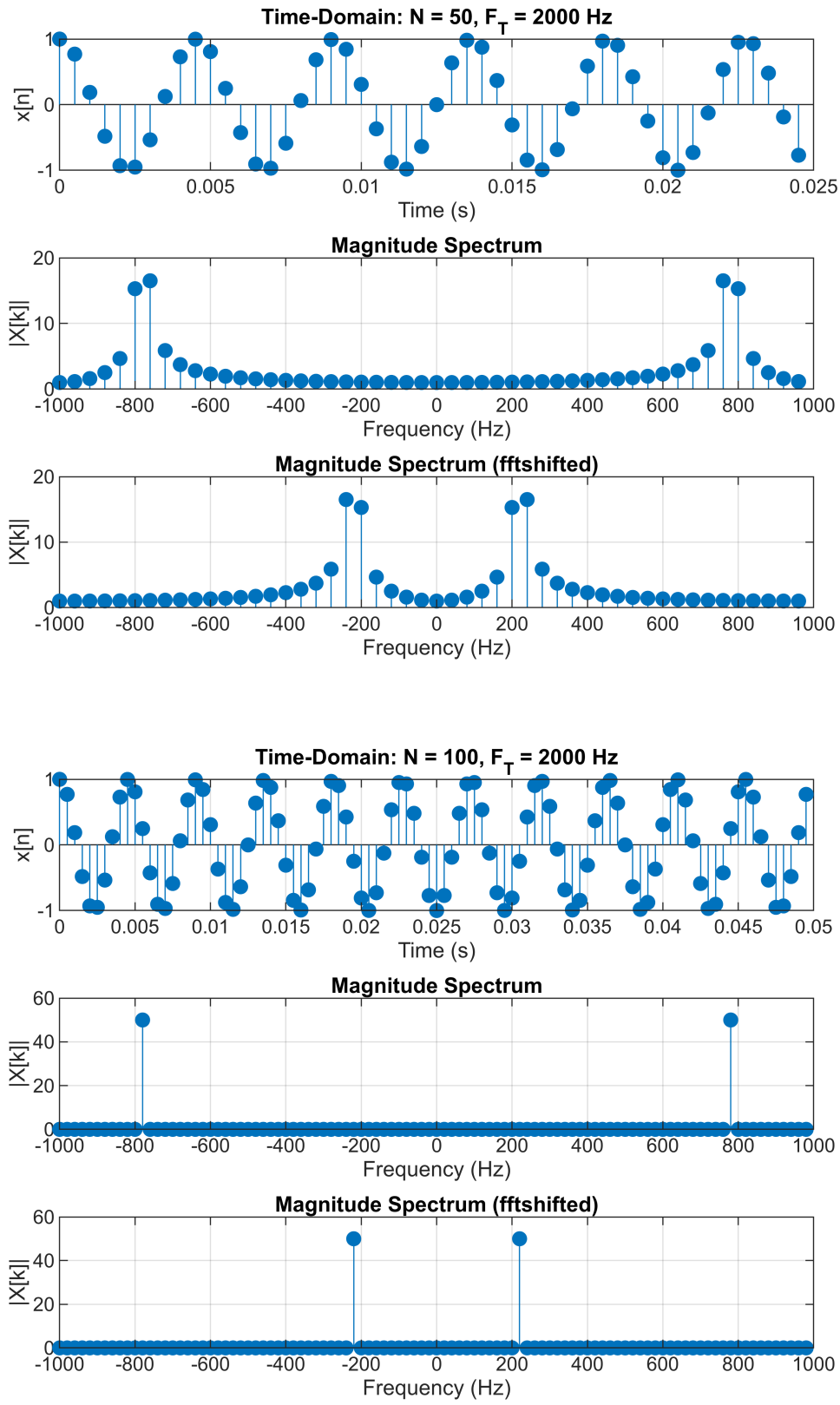
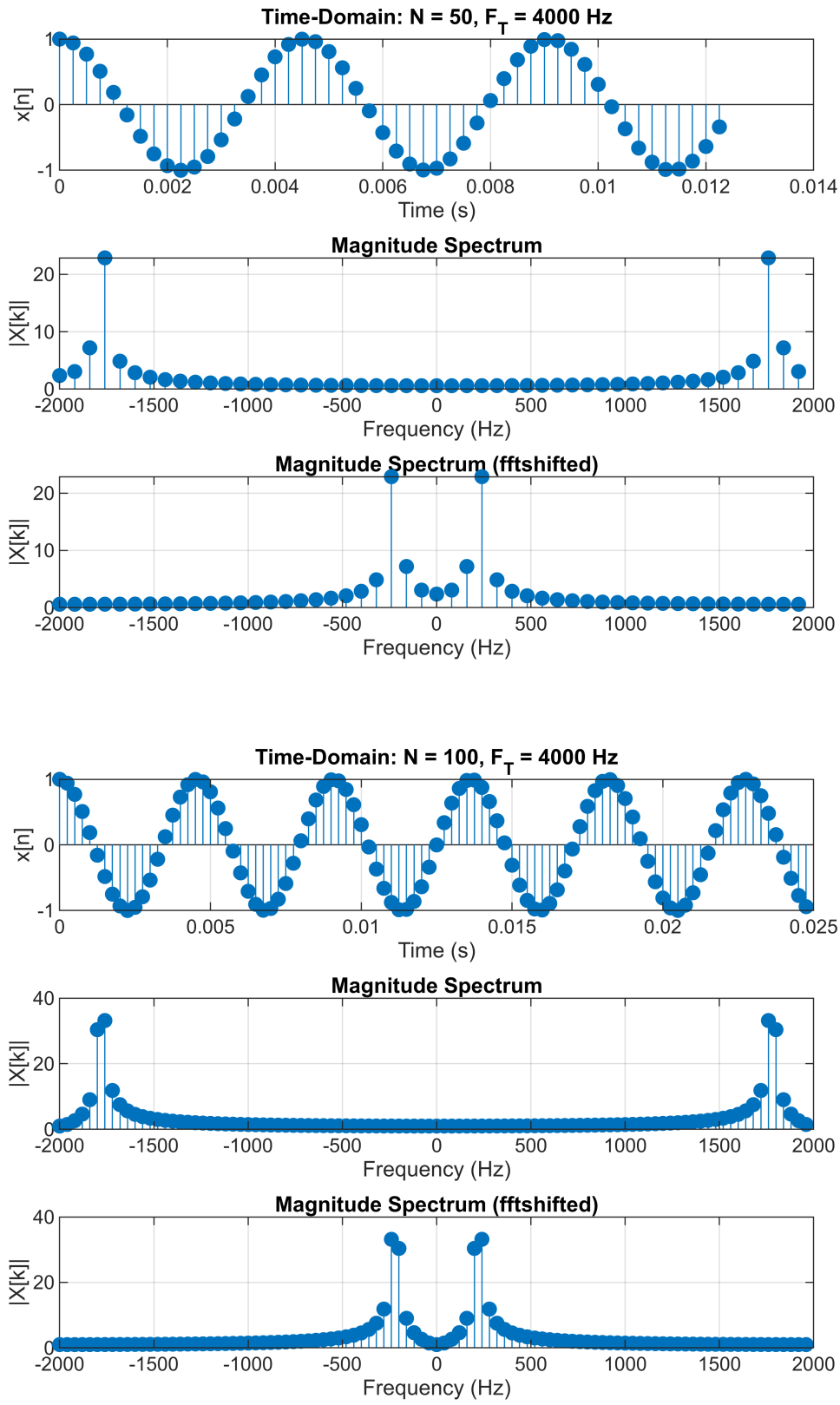
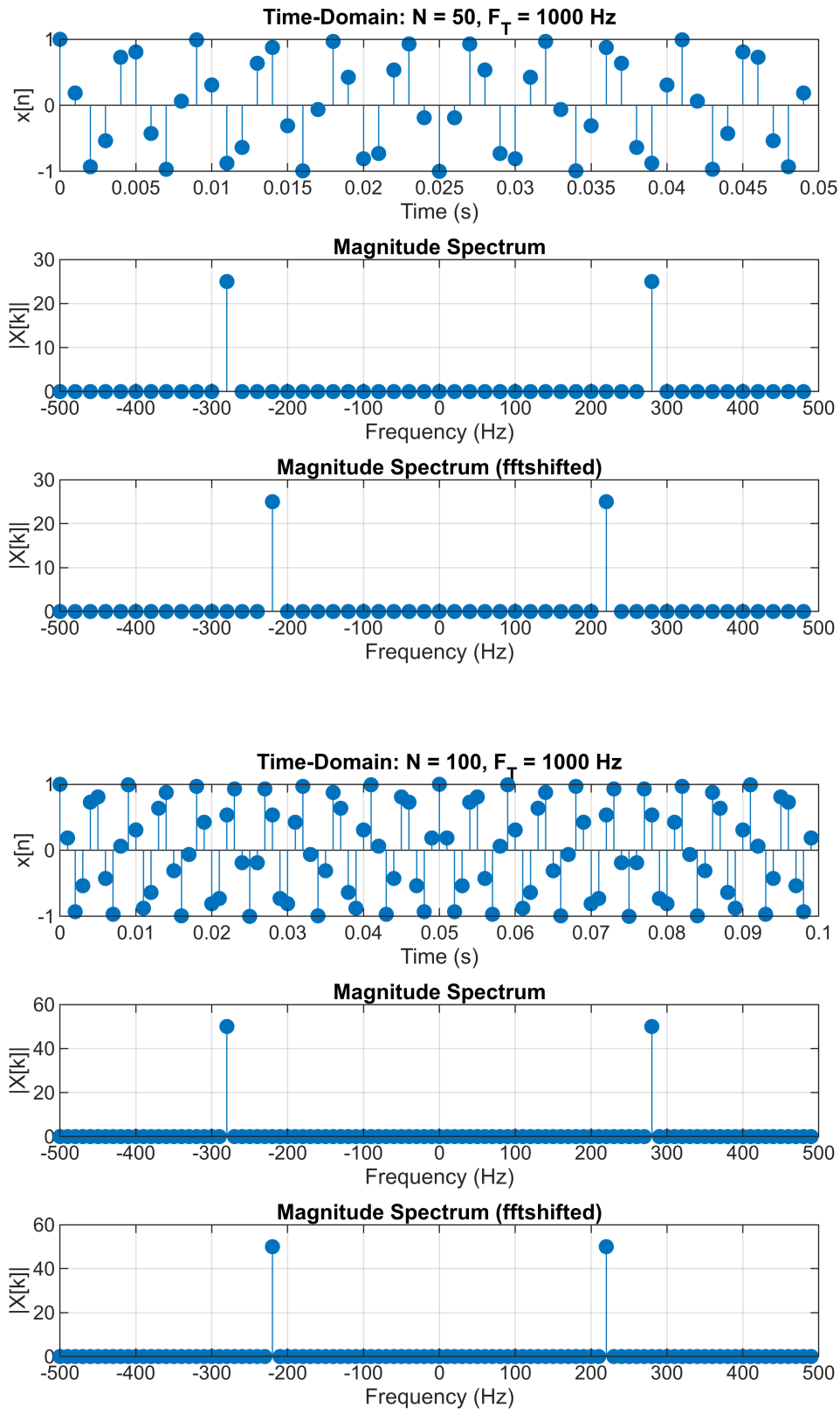
3. (10 points) The DFT is taken of N samples of $x(t) = \cos(2\pi(220)t)$ acquired with a sampling rate of F_T. For each of the following pairs of N and F_T : Use MATLAB to compute and stem plot the sample sequence with the horizontal axis labeled in units of time, not the sample index. Use MATLAB to compute and stem plot the magnitude of the DFT. Plot the DFT magnitude after using fftshift, and label the horizontal axis in units of Hz.

a.) N = 50 and $F_T = 2000$
b.) N = 100 and $F_T = 2000$
c.) N = 50 and $F_T = 4000$
d.) N = 100 and $F_T = 4000$
e.) N = 50 and $F_T = 1000$
f.) N = 100 and $F_T = 1000$

```
FT= [2000,2000,4000,4000,1000,1000];
N= [50,100,50,100,50,100];
for i=1:6
    T = 1 / FT(i);
    t = (0:N(i)-1) * T;
    x = cos(2*pi*220 * t);

    X = fft(x);
    Xshift = fftshift(X);
    f = (-N(i)/2:N(i)/2-1) * (FT(i) / N(i));

    figure;
    subplot(3,1,1);
    stem(t, x, 'filled');
    xlabel('Time (s)');
    ylabel('x[n]');
    title(sprintf('Time-Domain: N = %d, F_T = %d Hz', N(i), FT(i)));
    grid on;

    subplot(3,1,2);
    stem(f, abs(X), 'filled');
    xlabel('Frequency (Hz)');
    ylabel('|X[k]|');
    title('Magnitude Spectrum');
    grid on;

    subplot(3,1,3);
    stem(f, abs(Xshift), 'filled');
    xlabel('Frequency (Hz)');
    ylabel('|X[k]|');
    title('Magnitude Spectrum (fftshifted)');
    grid on;
end
```


Time-Domain: N = 50, F_T = 2000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)
Time-Domain: N = 100, F_T = 2000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)


Time-Domain: N = 50, F_T = 4000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)
Time-Domain: N = 100, F_T = 4000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)


Time-Domain: N = 50, F_T = 1000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)
Time-Domain: N = 100, F_T = 1000 Hz
x[n]
Time (s)
Magnitude Spectrum
|X[k]|
Frequency (Hz)
Magnitude Spectrum (fftshifted)
|X[k]|
Frequency (Hz)

5. (16 points) Use the DFT to explore harmonic effects of nonlinearities. For a linear system, if the input is a sinusoidal signal, the output will be a sinusoidal signal at the same frequency. For a nonlinear system, there can be frequencies in the output that are not found in the input.

a.) Create a vector x_1 [n] = cos(2*pi*(1/32)*n) for n=0 to 63. Compute the FFT and plot both the input vector and the magnitude of the FFT (after using *fftshift*). What is the magnitude of the largest component in the positive frequency range? To what frequency does it correspond? What is the second largest?

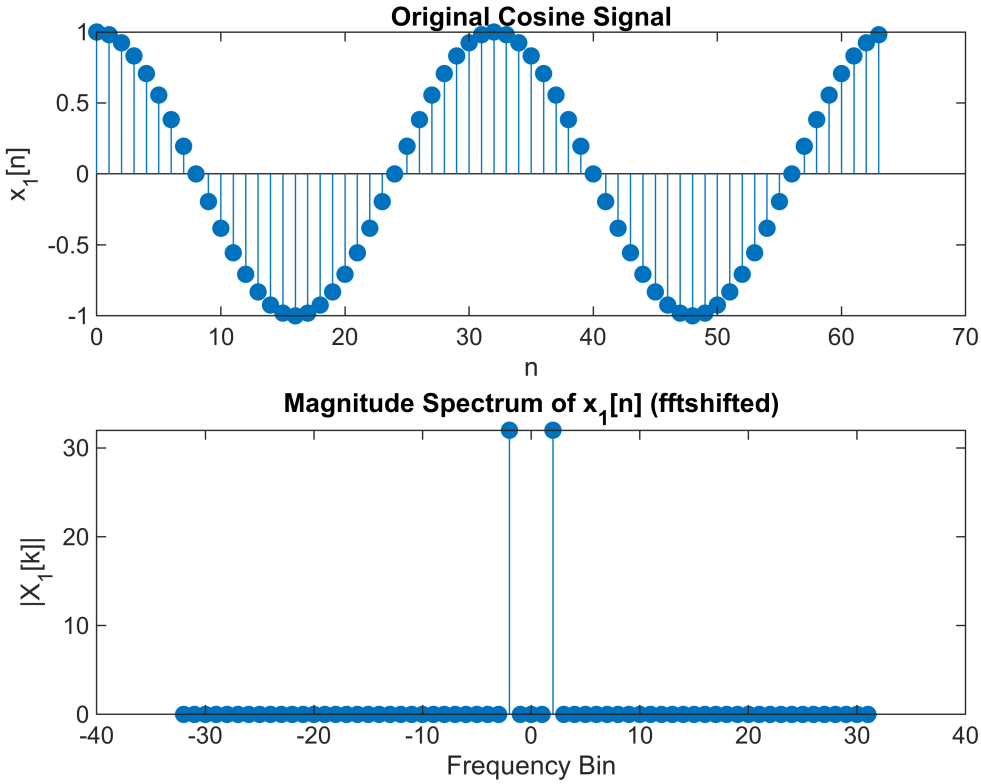
b.) Repeat (a) for $x_2[n] = x^3[n]$. Note the major harmonics and identify the frequencies of each.

c.) Repeat (b) for $x_3[n]$ = x[n] clipped at a magnitude of 0.9 with

$$x_3[n] = \begin{cases} x[n] & |x[n]| \le 0.9 \\ 0.9 & x[n] > 0.9 \\ -0.9 & x[n] < -0.9 \end{cases}$$

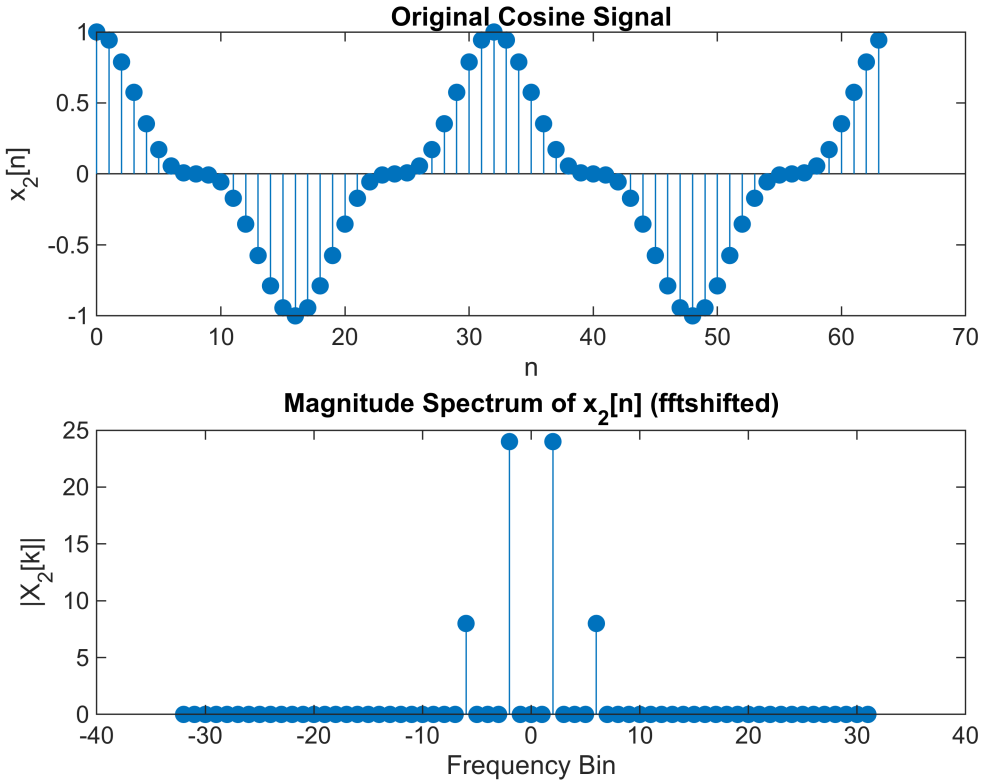
d.) Repeat (b) for x_4[n] = x[n]*(1 - 0.1*abs(x[n])).

```
n = 0:63;
x1 = cos(2*pi*(1/32)*n);

X1 = fftshift(fft(x1));
f = (-32:31);

figure;
subplot(2,1,1);
stem(n, x1, 'filled');
xlabel('n'); ylabel('x_1[n]');
title('Original Cosine Signal');

subplot(2,1,2);
stem(f, abs(X1), 'filled');
xlabel('Frequency Bin');
ylabel('|X_1[k]|');
title('Magnitude Spectrum of x_1[n] (fftshifted)');
```



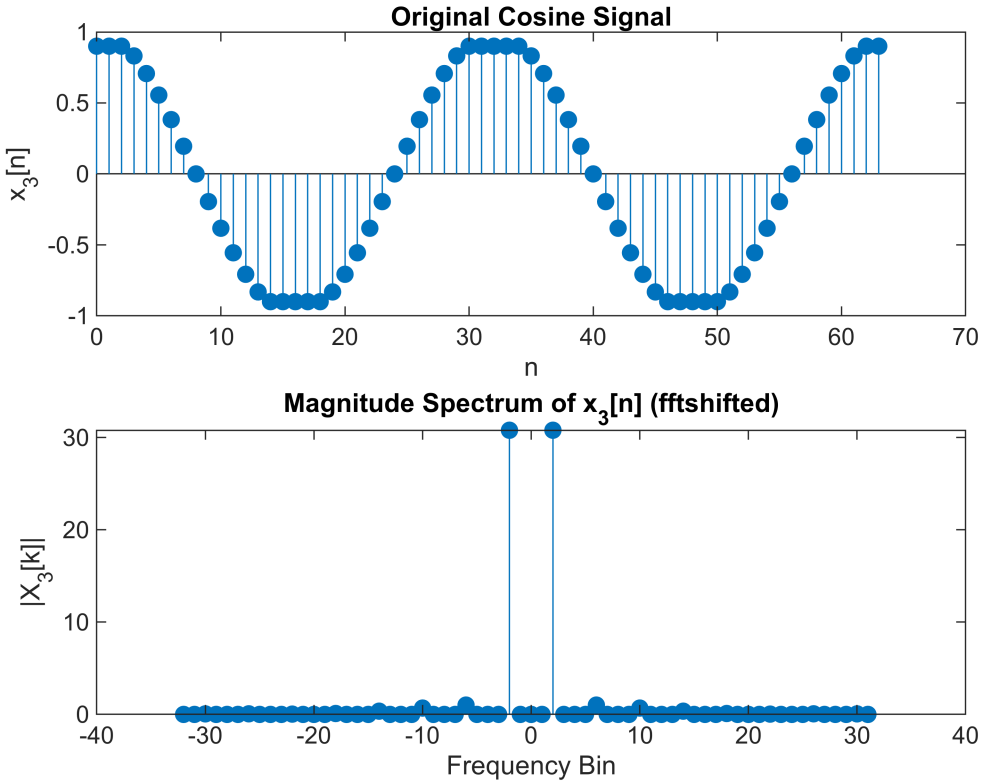
Magnitude of largest component is at bin 2 (freq=2Hz) and is 32. This is because the largest is N/2 when no leakage (which is what we see) and N=64. The second largest is also 32 at bin -2, but everything else is 0.

```
x2=x1.^3;

X2 = fftshift(fft(x2));
f = (-32:31);

figure;
subplot(2,1,1);
stem(n, x2, 'filled');
xlabel('n'); ylabel('x_2[n]');
title('Original Cosine Signal');

subplot(2,1,2);
stem(f, abs(X2), 'filled');
xlabel('Frequency Bin');
ylabel('|X_2[k]|');
title('Magnitude Spectrum of x_2[n] (fftshifted)');
```



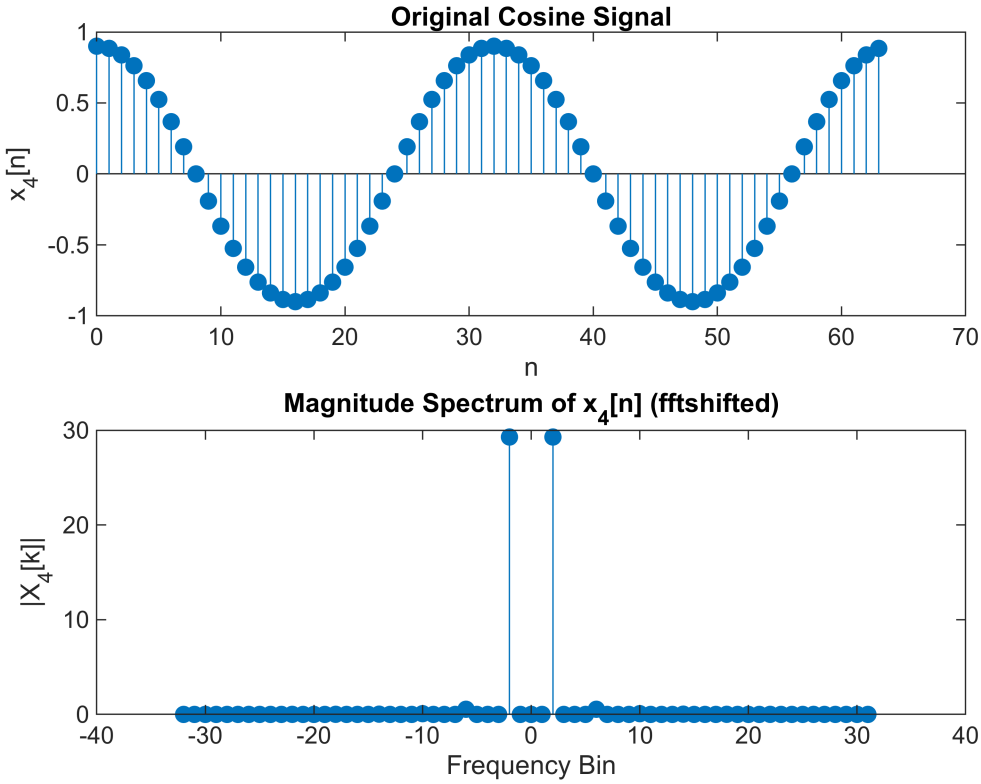
I see harmonics of amplitude 8 at -6 and 6 Hz, and 24 at -2 and 2Hz.

```
x3=x1;
x3(x1 > 0.9)  = 0.9;
x3(x1 < -0.9) = -0.9;

X3 = fftshift(fft(x3));
f = (-32:31);

figure;
subplot(2,1,1);
stem(n, x3, 'filled');
xlabel('n'); ylabel('x_3[n]');
title('Original Cosine Signal');

subplot(2,1,2);
stem(f, abs(X3), 'filled');
xlabel('Frequency Bin');
ylabel('|X_3[k]|');
title('Magnitude Spectrum of x_3[n] (fftshifted)');
```



Harmonics of decreasing value at bins +- 6,10,14,18,22,26,30. The major one is still at +-2.

```
x4= x1.*(1 - 0.1 * abs(x1));

X4 = fftshift(fft(x4));
f = (-32:31);

figure;
subplot(2,1,1);
stem(n, x4, 'filled');
xlabel('n'); ylabel('x_4[n]');
title('Original Cosine Signal');

subplot(2,1,2);
stem(f, abs(X4), 'filled');
xlabel('Frequency Bin');
ylabel('|X_4[k]|');
title('Magnitude Spectrum of x_4[n] (fftshifted)');
```



Harmonics at +-2 and small ones at +-6.